Deshik Reddy Putluru

Computational Mechanics | Finite Elements | FSI | Solver Development

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Professional Summary

Ph.D. candidate in computational mechanics specializing in finite-element formulations for continuum and fluid-structure interaction problems. Develops monolithic and mixed-dimensional models, stabilized discretizations, nonlinear solution strategies, and scalable preconditioners.

Education

Purdue University **Expected 2027**

Ph.D. in Mechanical Engineering, GPA: 3.87/4.00; Advisor: Prof. Hector Gomez *West Lafayette, IN*

Indian Institute of Technology Roorkee **2018 - 2022**

B.Tech. in Production and Industrial Engineering, with Honors; GPA: 8.98/10 *Roorkee, India*

Technical Expertise

Mechanics and discretization: Continuum mechanics, FEM, monolithic and mixed-dimensional FSI, ALE, membrane mechanics, hyperelastic solids, SAFE, reduced-order modeling.

Numerical methods: Variational formulations, residual-based VMS, SUPG, discontinuity capturing, Newton linearization, Krylov methods, saddle-point systems.

Solvers and HPC: GMRES, AMG/GAMG, block and field-split preconditioning, PETSc, MPI, parallel scientific computing.

Programming and software: C++, Python, MATLAB, FEniCS, OpenFOAM, COMSOL, Abaqus, ANSYS, ParaView.

Research Experience

Purdue University, Gomez Lab **May 2023 - Present**

Graduate Research Assistant - Computational Mechanics and Scientific Computing *West Lafayette, IN*

- Developed a mixed-dimensional finite-element model for spinal fluid-structure interaction, coupling a 2D Kirchhoff-Love dura membrane to a 3D hyperelastic solid and reducing cost by about $60\times$ relative to volumetric FSI.
- Implemented monolithic incompressible FSI solvers in FEniCS with ALE kinematics, residual-based VMS stabilization, Newton linearization, Krylov methods, and block-structured Jacobian assembly.
- Designed and benchmarked preconditioners for a double saddle-point FSI system. A GMRES/GAMG configuration matched a pinned FieldSplit solution at 3-4 times lower wall-clock cost while preserving flow and transport results.
- Developed the coupled momentum method as a reduced-cost alternative to ALE-FSI, avoiding mesh-motion overhead and enabling long-time transport simulations at roughly $10\times$ lower cost.
- Simulated injection-driven intrathecal transport at Peclet numbers near 10^7, using SUPG and discontinuity capturing to control instability and numerical diffusion in long-time drug-distribution predictions.

Selected Technical Projects

Hybrid SAFE analysis of Lamb waves in an elbow joint **Feb. 2021 - May 2021**

- Built a MATLAB-COMSOL implementation of the semi-analytical finite element method for guided Lamb-wave propagation through curved metallic structures.
- Performed parameter studies of how elbow geometry and material properties affect wave speed and mode characteristics.

Physics-constrained deep-learning surrogate for fluid flow **Jan. 2025 - May 2025**

- Implemented a label-free flow surrogate in JAX, Equinox, and Optax, vectorizing Navier-Stokes residual evaluation and automatic differentiation with `jit` and `vmap`.
- Validated predictions against analytical Poiseuille flow and CFD solutions for aneurysm and stenosis geometries.

Transient nucleate and film boiling with VOF **Jul. 2021 - Dec. 2021**

- Developed an OpenFOAM finite-volume solver using VOF and pressure/temperature-driven phase-change physics for a hot sphere immersed in a quiescent liquid.
- Resolved transient vapor-film dynamics across nucleate and film-boiling regimes and compared results with experiments and analytical predictions.

Shockwave propagation through a bifurcation **Aug. 2021 - Nov. 2021**

- Recreated a bifurcating shock-tube experiment in OpenFOAM, identified realistic boundary conditions, and validated the solution strategy against measurements.
- Performed a geometric study of how bifurcation angle changes junction overpressure and local flow structure.

Publications

- **Putluru, D. R.**, Tepole, A. B., and Gomez, H. "Mixed-dimensional fluid-structure interaction simulations reveal key mechanisms of cerebrospinal fluid dynamics in the spinal canal." *Fluids and Barriers of the CNS* 22, 81 (2025). doi:10.1186/s12987-025-00691-4.
- **Putluru, D. R.** and Gomez, H. "On the role of tissue compliance in intrathecal drug transport." In preparation (2026).

Honors and Research Communication

- Professional Development and Innovation Award for Bachelor Thesis Project (2022); first place in ASME human-powered vehicle design (2020); national top 1 percent in NSEJS (2015).
- Poster presenter at the Purdue One Health and Well-Being Research Event (2026); LPRC panelist; led a SolidWorks/ANSYS workshop for more than 150 students at IIT Roorkee.